

Case 3: ALFWorld Examine Task (Pencil ⊕ Desklamp → Success)

Task: Look at pencil under the desklamp.

Core Skills	[loo_001] Seek lamp surfaces (Desk/Sidetable); [loo_002] Switch lamp on; [exa_001] Pair objects early (Target & Tool co-location).
Mistakes Avoided	[err_003] Ignoring reachable targets; [loo_005] Preventing action loops (Toggle only once).

Agent Trajectory: High-Efficiency Co-location Handling

Step 1	Obs: Mid-room. Visible: desk 1/2, drawer 1-3, sidetable 1, etc. Reasoning: [loo_001] Desk lamps usually sit on desks. Prioritizing desk 1 to find both pencil and lamp in one go. Action: go to desk 1
Step 2	Obs: On desk 1: alarmclock 1, desk lamp 1, pencil 1. Reasoning: [exa_001] Successful co-location! Both objects are present. [exa_002] Pick up target first before interacting with the tool. Action: take pencil 1 from desk 1
Step 3	Obs: Holding pencil 1. Reasoning: [loo_002] Toggle the lamp state to satisfy the "under light" condition. [exa_004] Single toggle rule ensures no repeated actions. Action: use desk lamp 1

Result: SUCCESS (Optimal 3 Steps) — Demonstrates superior spatial reasoning and zero-waste navigation. Key Skills: [exa_001] (Object Pairing), [loo_001] (Spatial Priors).